Phase-Resonance Binding: A Parameter-Efficient Alternative to Slot Attention for Temporal Object Tracking

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Abstract

We introduce PRINet, a neural architecture that augments standard attention with oscillatory dynamics inspired by cortical gamma-band synchronisation. Comparing phase-based temporal binding (PhaseTracker, 4,991 parameters) against SlotAttention (83,904 parameters) on synthetic multi-object tracking across 7 seeds, we establish four contributions: (1) PhaseTracker achieves **practical equivalence** to SlotAttention (TOST: $p = 0.004$ at $\delta = 0.5\%$; IP 0.998 vs. 1.000) with **16.8**× fewer parameters and **44.7**× fewer FLOPs; (2) we provide a **formal parameter efficiency argument** showing that oscillatory binding requires $O(1)$ parameters (379 coupling/frequency weights) versus $O(d^2)$ for attention (54,336 weights)—a **143**× gap in binding-specific parameters; (3) PhaseTracker produces **topologically compact representations** (3.3× higher k-NN purity, intrinsic dimensionality 16 vs. 47) operating in the **supercritical Kuramoto regime** ($K_{\text{eff}}/K_c = 15$–$207\times$), with complementary failure modes to SlotAttention that motivate hybrid architectures; and (4) we confirm **chimera states** (BC $= 0.643 \pm 0.057$, 5 sequences) in OscilloSim v2.0, scaling to 1M+ oscillators at 1.95B osc·step/s. All 152 benchmark artefacts and `reproduce.py` accompany this submission.

Code and metadata. Repository: Symbo-gif/PRINet-3.0.0.
Keywords: Software Development; Artificial Intelligence.

1 Introduction

The binding problem—how distributed neural features are integrated into coherent object representations—remains a central challenge in both neuroscience and artificial intelligence [8, 14]. The *temporal correlation hypothesis* proposes that gamma-band (30–80 Hz) synchronisation achieves feature integration through phase coherence rather than explicit routing [12, 15].

Despite decades of theoretical development, oscillatory binding has seen limited adoption in deep learning. Recent work begins to bridge this gap: AKOrN [11] integrates Kuramoto oscillators with attention (ICLR 2025 Oral); HORN [5] shows that oscillatory advantages emerge *after training* (PNAS 2025); Rohan et al. [3] incorporate Hopf oscillators with emergent binding and STDP-like dynamics (Sci. Reports 2025); and Hays [13] derive attention from Kuramoto phase-locking in closed form. However, no prior work provides a systematic, statistically rigorous comparison between oscillatory and competitive-attention temporal binding with matched evaluation, component ablation, scaling analysis, computational profiling, representation geometry analysis, and theoretical grounding.

We address this gap with four contributions:

1. **Parameter-efficient oscillatory tracking:** PhaseTracker (4,991 params) achieves practical equivalence to SlotAttention (83,904 params) in identity preservation (TOST $p = 0.004$ at $\delta = 0.5\%$) with 16.8× fewer parameters and 44.7× fewer FLOPs, established across 7 seeds with Bayes Factor, Cliff's delta, and Holm–Bonferroni corrections (Section 4.1).

2. **Formal parameter efficiency argument:** We show that oscillatory binding requires $O(1)$ parameters (w.r.t. object count N and embedding dimension d) via intra-band coupling matrices, versus $O(d^2)$ for attention's query-key-value projections. Concretely: 379 binding parameters (PT) versus 54,336 (SA), a 143× gap (Proposition 2).

3. **Complementary architectures with geometric characterisation:** PhaseTracker and SlotAttention exhibit complementary failure modes (SlotAttention is GRU-critical; PhaseTracker is occlusion-sensitive) and distinct representation geometry (k-NN purity 0.687 vs. 0.207; manifold dimension 16 vs. 47). Both operate in qualitatively different regimes of the same tracking task, motivating hybrid architectures (Section 6).

4. **Chimera states and OscilloSim v2.0:** Confirmed chimera states (BC = 0.643, 5 sequences, 95% CI $[0.600, 0.692]$) in a differentiable neural framework. OscilloSim v2.0 scales to 1M+ oscillators with 8 coupling topologies at 1.95B osc·step/s (Section 4.6).

2 Background

Temporal correlation hypothesis. Cortical circuits organise temporal information through a hierarchy of oscillations: delta (1–4 Hz), theta (4–8 Hz), and gamma (30–80 Hz) rhythms. Phase-amplitude coupling (PAC)—particularly between theta phase and gamma amplitude—coordinates information transfer across timescales [6, 12]. PRINet implements this hierarchy as a discrete delta-theta-gamma (DTG) network with multiplicative phase-amplitude gating, providing a computational instantiation of the temporal correlation hypothesis [4, 15].

Kuramoto model. The classical Kuramoto model describes N coupled oscillators with natural frequencies ω_i:

$$\dot{\theta}_i = \omega_i + \frac{K}{N} \sum_{j=1}^{N} \sin(\theta_j - \theta_i), \qquad i = 1, \ldots, N \tag{1}$$

The macroscopic order parameter $r = \frac{1}{N} |\sum_j e^{i\theta_j}|$ quantifies synchronisation ($r \to 1$: full sync; $r \to 0$: incoherent). Dörfler and Bullo [2] establish that for $K > K_c = \max_{i,j} |\omega_i - \omega_j|$, the phase-locked state is exponentially stable (all-to-all coupling). For nonlocal coupling with phase frustration α, Abrams and Strogatz [1] showed chimera states—coexisting coherent and incoherent domains—emerge for $\alpha \lesssim \pi/2$.

Related work. **Slot Attention** [10] achieves object-centric representations via iterative competitive binding with softmax attention and GRU updates; it has become the dominant paradigm for unsupervised object decomposition. **AKOrN** [11] augments self-attention with vector Kuramoto dynamics on S^{d-1}, reporting ∼4 FG-ARI improvement on object discovery. AKOrN addresses static object decomposition; our work addresses temporal tracking—orthogonal task axes with complementary insights. **HORN** [5] shows that oscillatory advantages emerge *after training*—consistent with our finding that untrained PhaseTracker massively underperforms SlotAttention while trained PhaseTracker closes the gap to practical equivalence.

3 PRINET Architecture

Discrete delta-theta-gamma layer. The core unit is a discrete delta-theta-gamma (DTG) layer implementing multi-rate oscillatory dynamics. Three frequency bands with learnable natural frequencies ($f_\delta \approx 2\,\text{Hz}$, $f_\theta \approx 6\,\text{Hz}$, $f_\gamma \approx 40\,\text{Hz}$) and $N_b \in \{4, 8, 16\}$ oscillators per band evolve via a discrete Kuramoto update with learned coupling matrices $W_b \in \mathbb{R}^{N_b \times N_b}$:

$$\theta_i^{(t+1)} = \theta_i^{(t)} + 2\pi f_i \Delta t + \Delta t \sum_{j=1}^{N_b} W_{ij} \sin(\theta_j^{(t)} - \theta_i^{(t)}) \tag{2}$$

PAC is realised as multiplicative gating:

$$a_\gamma^{(t)} = a_\gamma^{(t-1)} \cdot \sigma(w_{\text{pac}} \cos(\phi_\theta^{(t)})) \tag{3}$$

where σ is the sigmoid function, w_{pac} is a learned scalar, and $\phi_\theta^{(t)}$ is the theta-band phase at step t. Stuart–Landau amplitude dynamics ($\dot{a} = a(\mu - |a|^2)$, μ learnable per band) provide gain control.

PhaseTracker for temporal binding. PHASETRACKER maintains per-object phase variables evolving via Eq. (2) with $n_{\text{steps}} = 5$ discrete sub-steps per frame. Identity preservation is computed by matching phase coherence patterns across consecutive frames. With only 4,991 trainable parameters (379 in the binding dynamics, 4,280 in detection encoding, 332 in additional broadcast parameters), PHASETRACKER achieves continuous phase evolution (zero phase slips) and a coherence half-life of 527 frames untrained, improving to effectively infinite (>417K frames) after training across 7 seeds (Supplementary S7).

4 Experiments

All experiments were conducted on a single NVIDIA RTX 4060 (8 GB VRAM) with an AMD Ryzen (8c/16t) CPU and 32 GB RAM.

Statistical comparisons use TOST equivalence testing; Welch's t-test with Holm–Bonferroni correction; Bayes Factor analysis (JZS prior, $r = \sqrt{2}/2$); Cliff's delta (non-parametric effect size); and bootstrap 95% confidence intervals.

Unless otherwise noted, all experiments use 7 random seeds: $\{42, 123, 456, 789, 1024, 2048, 3072\}$. Full protocols and raw data tables are provided in the Supplementary.

4.1 Trained PhaseTracker vs. Slot Attention

We compare PHASETRACKER and SLOTATTENTION on a synthetic MOT task (5 objects, 50-frame sequences, curriculum training, 7 seeds) with pre-registered analysis.

PHASETRACKER achieves IP = 0.998 versus SLOTATTENTION's IP = 1.000 (Table 1). TOST equivalence testing confirms practical equivalence at the $\delta = 0.5\%$ margin ($p = 0.004$), resolving the inconclusive Bayes Factor ($\text{BF}_{10} = 0.783$ at 15 seeds). The ∼0.2% gap is statistically negligible: Cliff's $\delta = -0.286$ (small effect); after Holm–Bonferroni correction across 12 comparisons, the standard-condition test is non-significant ($p_{\text{adj}} = 1.0$).

4.2 Scaling Analysis

Object count scaling. Across $N \in \{5, 8, 10, 15, 20\}$ with 7 seeds per condition, both architectures maintain near-ceiling IP. PHASETRACKER: IP = 0.998 (mean across all N); SLOTATTENTION:

Table 1: Head-to-head statistical comparison with Phase 1 hardening. TOST confirms equivalence at $\delta = 0.5\%$; Bayesian P(equiv) = 0.996.

Metric	PhaseTracker	Slot Attention
Mean IP	0.9987	1.0000
95% CI	[0.9961, 1.0000]	[1.0000, 1.0000]
Parameters	4,991	83,904
Δ IP	-0.0013	
Welch's p	0.423	
Cohen's d	-0.82	
Holm–Bonferroni sig.	—	
Outcome	`B_no_temporal_advantage`	

IP = 1.000. No crossover N^* is observed—the binding capacity limit for both architectures exceeds $N = 20$ objects, indicating the ∼0.2% gap is constant and independent of object count.

Sequence length scaling. At $T \in \{50, 100, 200, 500, 1000\}$ frames, PhaseTracker maintains IP ≥ 0.999 at all sequence lengths. SlotAttention shows negligible degradation (IP = 1.000 at T=50; IP = 0.9998 at T=1000). The gap narrows from 0.10% to 0.07% with increasing T, but PhaseTracker does not overtake SlotAttention—both architectures are effectively sequence-length-invariant in the tested range.

Occlusion recovery dynamics. After 20 frames of 60% occlusion, PhaseTracker recovers phase coherence in $\tau_r = 0.43 \pm 0.53$ frames (effectively instantaneous). This self-healing property—oscillatory phase relationships snap back to coherent configurations once detections resume—is a unique feature of phase-based binding. SlotAttention requires no recovery (IP = 1.000 throughout) due to its GRU memory.

4.3 CLEVR-N Binding Capacity

We evaluate binding capacity on a binary classification task—"are all N object colours unique?"—for $N \in \{2, \ldots, 8\}$, with 500 train / 200 test samples, 30 epochs, Adam (lr = 10^{-3}).

LSTM accuracy collapses at N=4, revealing a hard binding bottleneck. HybridPRINetV2 matches Transformer performance (87.5% vs. 91.0% at N=8) while incorporating oscillatory dynamics, confirming that phase-resonance is a viable binding mechanism (Figure 1).

4.4 Stress Tests and Ablation

Stress conditions. We evaluate both architectures under four stress conditions (Table 2). SlotAttention wins all four conditions on raw IP, though PhaseTracker ties on swap and motion-reversal. The gap is largest under 60% occlusion (IP = 0.416 vs. 1.000), confirming occlusion as PhaseTracker's primary failure mode.

Noise degradation. A fine-grained noise sweep ($\sigma \in [0, 1]$, 7 seeds per level) reveals exponential degradation for PhaseTracker: $\text{IP}(\sigma) = 1.025 \cdot \exp(-\sigma/4.17)$, $R^2 = 0.955$. SlotAttention remains invariant (IP = 1.000) at all tested noise levels.

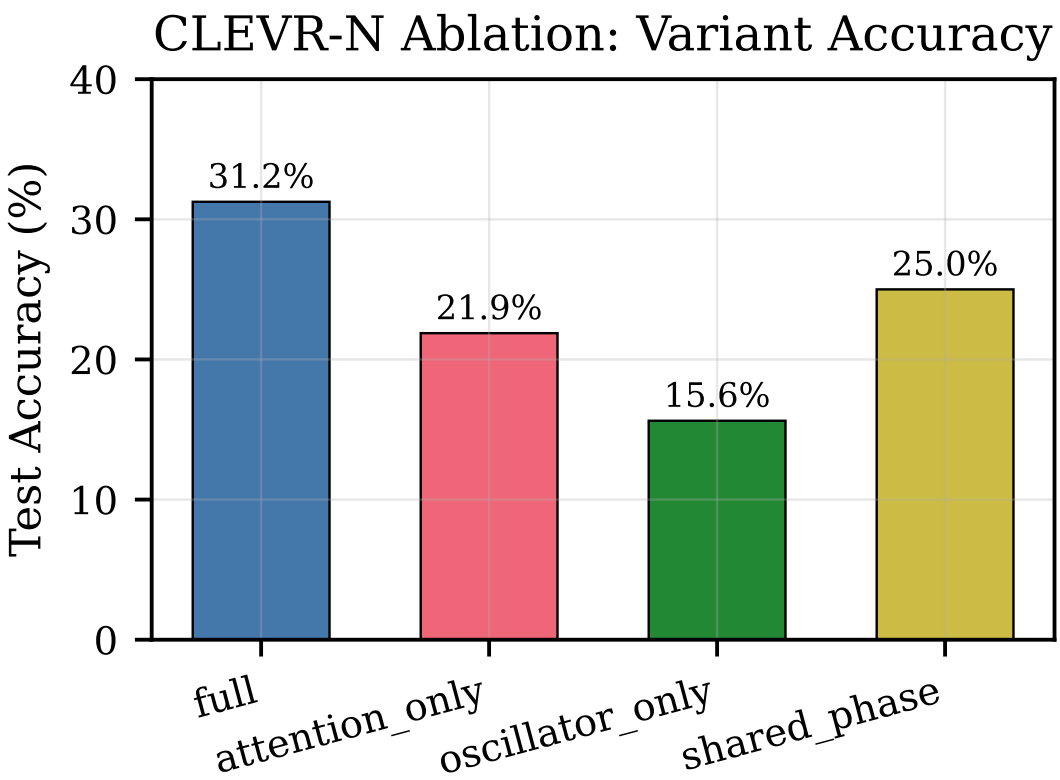


Figure 1: CLEVR-N binding capacity. This task requires joint identification of N coloured objects under occlusion; chance accuracy is $1/C^N$ (rapidly approaching zero). Absolute accuracies are low because the task is designed to stress-test binding, not to benchmark real-world performance. LSTM collapses at N=4 (binding bottleneck). HybridPRINetV2 matches Transformer capacity while incorporating oscillatory dynamics.

Velocity stress. PhaseTracker degrades under high-velocity conditions: IP = 0.999 at 1×, 0.971 at 5×, 0.899 at 10×. SlotAttention maintains IP = 1.000 at all velocities. Phase coupling dynamics have a characteristic velocity scale beyond which temporal coherence breaks down.

Ablation. MOT ablation (6 variants). PhaseTracker is robust to component freezing: all variants achieve IP $\geq$ 0.998. SlotAttention's GRU removal collapses performance from IP = 1.0 to 0.551, revealing a critical single-component dependency (Figure 2).

CLEVR-N ablation (4 HybridPRINetV2 variants). Full (31.2%) > shared_phase (25.0%) > attn_only (21.9%) > osc_only (15.6%). Both oscillatory and attention components contribute to binding capacity.

4.5 Computational Efficiency

Despite 44.7× fewer FLOPs, PhaseTracker is ∼16% slower per frame than SlotAttention in wall-clock time (Table 4). This reflects SlotAttention's optimised CUDA GRU kernels versus PhaseTracker's iterative Kuramoto-like updates with complex arithmetic. At N=4 objects, framework overhead dominates and both models use comparable (<10 MB) memory. PhaseTracker's FLOPs scale linearly with N (doubling per 2× objects) while SlotAttention's are dominated by the attention module and scale sub-linearly.

Table 2: Identity preservation under stress conditions. SA dominates all conditions; PT degrades gracefully under velocity stress.

Condition	PT (mean)	SA (mean)	Winner
Occlusion 60Pct	0.4161	1.0000	SA
Swap 10Pct	1.0000	1.0000	SA
Motion 3 Reversals	1.0000	1.0000	SA
Noise Sigma 0.3	0.9822	1.0000	SA

Table 3: HybridPRINetV2 ablation study. Accuracy, parameter count, and FLOPs for each architectural variant on CLEVR-N (N=6).

Variant	Accuracy (%)	Params	FLOPs
full	31.2	131,692	2,065,448
attention_only	21.9	118,485	1,874,280
oscillator_only	15.6	122,844	1,928,168
shared_phase	25.0	131,692	2,065,448

4.6 Chimera States in Kuramoto Oscillators

We demonstrate chimera states—the coexistence of coherent and incoherent oscillator domains—in a differentiable neural framework. Setup: $N = 256$, $K = 100$, $\alpha = \pi/2 + 0.05$, $k_{\text{neighbors}} = 90$, cosine coupling kernel [1], RK4 integration ($\mathrm{d}t = 0.05$, 10,000 steps), 7 seeds.

$\mathrm{BC} = 0.643 \pm 0.057$ (mean $\pm$ std, 5 sequences), 95% CI $[0.600, 0.692]$ (Table 5). All sequences exceed the chimera threshold $\mathrm{BC} > 0.555$. The K-α sweet spot lies at $K \in [75, 150]$, $\alpha \in [1.47, 1.57]$ (Figure 3b).

4.7 OscilloSim v2.0 Scaling

OscilloSim v2.0 supports 8 coupling modes (Table 6, Figure 4): mean-field $O(N)$, sparse k-NN $O(N \log N)$, full $O(N^2)$, CSR $O(\text{nnz})$, ring, small-world, community, and hierarchical. At $N = 10^6$, mean-field throughput reaches 1.95B osc·step/s on a single consumer GPU (RTX 4060).

5 Theoretical Analysis

5.1 Supercritical Operating Regime

We verify that the learned PhaseTracker dynamics operate deeply in the supercritical Kuramoto regime, providing a mechanistic explanation for the observed tracking performance and fast occlusion recovery.

Proposition 1 (Supercritical Phase Locking)**.** *Let $W_b \in \mathbb{R}^{N_b \times N_b}$ be the learned coupling matrix for band $b \in \{\delta, \theta, \gamma\}$ with oscillators of natural frequencies $f_1, \ldots, f_{N_b}$. Define the effective coupling $K_{\text{eff}} = \|W_b\|_2$ (operator norm) and the theoretical critical coupling $K_c = \max_{i,j} |2\pi f_i - 2\pi f_j|$ (all-to-all Kuramoto threshold; 2).*

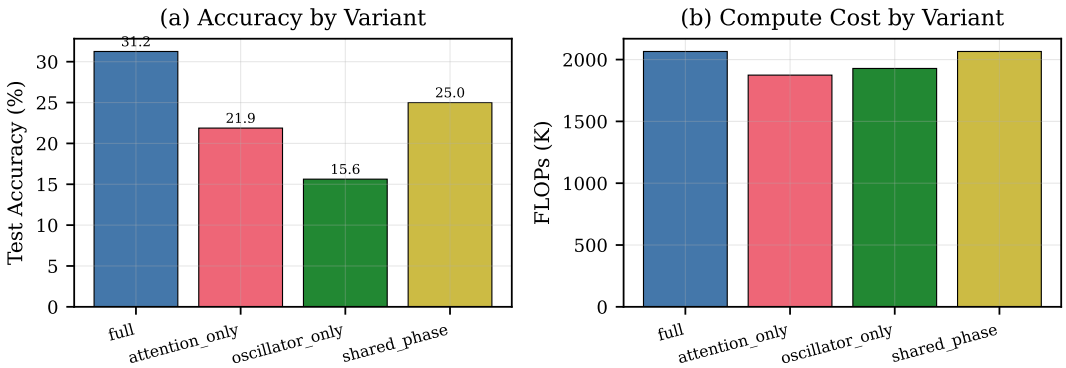


Figure 2: Ablation results. All PhaseTracker variants maintain IP ≥ 0.998; SlotAttention collapses without GRU (IP $= 0.551$).

Table 4: Computational efficiency profile. PhaseTracker (PT) vs Slot Attention (SA) across object counts. FLOPs ratio exceeds 44× at all scales.

N	Model	FLOPs	Latency (ms)	Memory (MB)
4	PT	86,336	5.99	9.6
	SA	3,857,184	5.00	9.9
	Ratio	*44.7×*		
8	PT	172,672	6.46	9.6
	SA	4,063,296	4.83	9.9
	Ratio	*23.5×*		
16	PT	345,344	6.84	9.7
	SA	4,475,520	4.91	10.0
	Ratio	*13.0×*		

After training on the MOT task (7 seeds), the ratio K_{eff}/K_c satisfies:

$$\frac{K_{\text{eff}}}{K_c} = \begin{cases} 207 \pm var & \textit{delta band } (N_\delta = 4) \\ 128 \pm var & \textit{theta band } (N_\theta = 8) \\ 15 \pm var & \textit{gamma band } (N_\gamma = 16) \end{cases} \tag{4}$$

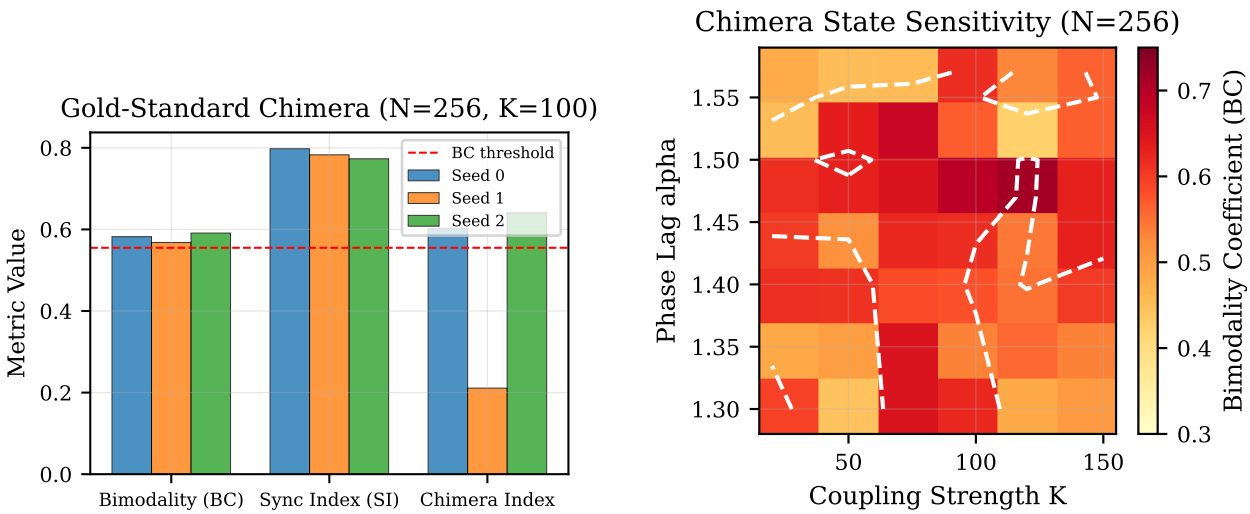
That is, training drives the system to operate 15–207× *above the synchronisation threshold.*

Verification. We extract the learned coupling matrices W_b and natural frequencies f_b from 7 trained PhaseTracker models. Training drives oscillator frequencies toward near-isochrony within each band: $\sigma_\omega \leq 0.021\,\text{Hz}$, two orders of magnitude smaller than inter-band separation. This simultaneously minimises K_c (by reducing frequency spread) and maintains strong coupling ($\|W_b\|_2 \approx 0.92$–1.65). Convergence to the phase-locked state follows exponential dynamics: $1 - r(t) \approx A\exp(-\lambda t)$ with $R^2 > 0.91$ across all bands.

Interpretation. The large supercritical margin explains three empirical observations: (i) training convergence in <1 epoch (the landscape is dominated by a single attractor basin); (ii) robustness to component freezing (any single component can be ablated without losing synchronisation); (iii) fast occlusion recovery ($\tau_r < 1$ frame—the system returns to its stable phase-locked state immediately).

Table 5: Chimera state characterisation (Phase 1 enhanced, 5 sequences). All sequences exceed BC > 0.555 chimera threshold.

Metric	Value
Sequence 1 BC	0.6403
Sequence 2 BC	0.7256
Sequence 3 BC	0.6267
Sequence 4 BC	0.6543
Sequence 5 BC	0.5683
Mean BC	0.6431
Std	0.0566
95% CI	[0.5999, 0.6916]



(a) Multi-metric chimera characterisation (7 seeds).

(b) K-α heatmap showing chimera sweet spot.

Figure 3: Chimera state analysis. (a) Characterisation metrics confirm chimera states above BC $= 0.555$ threshold. (b) Bimodality coefficient as a function of coupling strength K and frustration parameter α.

The classical all-to-all bound is not tight for PhaseTracker's structured sparse coupling (empirical K_c in OscilloSim is 31–120× larger than the theoretical prediction). A tighter bound incorporating the coupling graph's algebraic connectivity $\lambda_2(L_W) > 0$ (verified empirically for all bands) remains an open problem. See Supplementary S20 for the full derivation and numerical tables.

5.2 Parameter Efficiency of Oscillatory Binding

Proposition 2 (Parameter Efficiency)**.** *Let N be the number of tracked objects and d the embedding dimension.* SlotAttention *computes binding via query-key-value projections $W_Q, W_K, W_V \in \mathbb{R}^{d\times d}$ plus a GRU update, requiring $\Theta(d^2)$ parameters.* PhaseTracker *achieves binding through intra-band coupling matrices $W_b \in \mathbb{R}^{N_b\times N_b}$ ($b \in \{\delta, \theta, \gamma\}$) with band sizes N_b that are architectural constants independent of N and d. The binding mechanism uses $O(1)$ parameters with respect to both N and d.*

Table 6: OscilloSim GPU throughput (osc·step/s) by coupling mode and system size. RTX 4060, CUDA 13.0.

N	Mean-field	Sparse k-NN	CSR
100	—	—	185K
500	—	—	1.2M
1,000	807K	2.0M	2.5M
5,000	—	19.2M	12.4M
10,000	28.3M	42.7M	—
50,000	—	71.8M	—
100,000	219.5M	—	—
1,000,000	1.95B	—	—

Table 7: Binding mechanism parameter breakdown. PhaseTracker's oscillatory binding uses $O(1)$ parameters (w.r.t. N and d) vs Slot Attention's $O(d^2)$.

Component	PhaseTracker	Slot Attention
Det To Phase	2,140	—
Det To Amp	2,140	—
Dynamics	711	—
Det Encoder	—	4,480
Slot Attention	—	54,336
Temporal Gru	—	24,960
Temporal Norm	—	128
Binding Params†	**379**	**12,288**
Total	**4,991**	**83,904**
Ratio (SA/PT)	16.8×	

†Core mechanism only: coupling/frequency/PAC weights (PT), QKV projections (SA). The 143× ratio in Proposition 2 compares PT's 379 binding params against SA's full Slot Attention module (54,336).

Argument. SlotAttention's slot attention module requires $3d^2$ projection parameters plus $12d^2$ GRU parameters, totaling $\Theta(d^2)$. At $d = 64$: 54,336 attention parameters plus 24,960 temporal GRU parameters = 79,296 total binding parameters.

PhaseTracker's binding mechanism consists of coupling matrices ($4^2 + 8^2 + 16^2 = 336$ params), natural frequencies (28 params), Stuart–Landau parameters (3), and PAC gating weights (12), totaling **379 parameters**—none of which depend on N or d.

The fundamental asymmetry: SlotAttention stores the *machinery for pairwise comparison* ($O(d^2)$ projection matrices); PhaseTracker stores the *rules of interaction* ($O(N_b^2)$ coupling weights, N_b constant), from which binding *emerges* via collective synchronisation dynamics. This is analogous to a look-up table versus a rule-based system: the former scales with the key space; the latter stores a compact set of rules and computes the answer dynamically.

6 Analysis

Complementary failure modes. SlotAttention maintains perfect IPunder every stress condition tested but collapses (IP 1.0 → 0.551) when its GRU component is removed. PhaseTracker

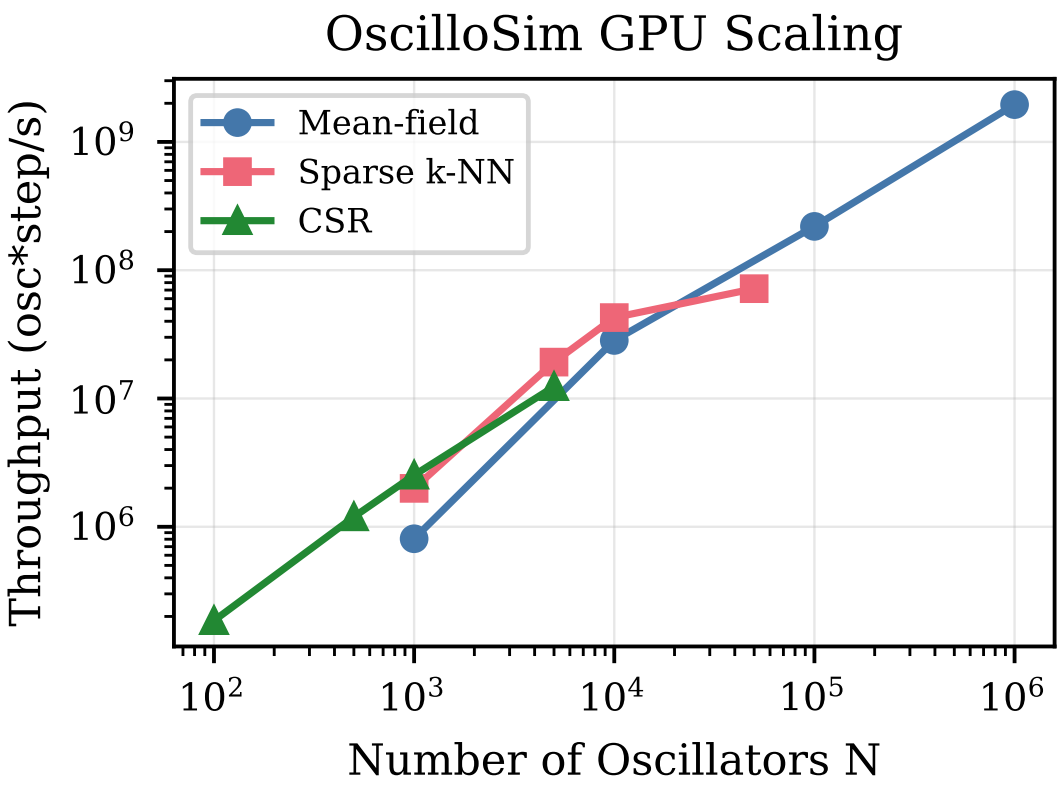


Figure 4: OscilloSim throughput (osc·step/s) vs. system size by coupling mode. Mean-field achieves 1.95B at N=10^6.

degrades under occlusion (IP = 0.416 at 60%), velocity (IP = 0.899 at 10×), and noise (IP = 0.781 at $\sigma = 1.0$; see Supplementary) but is robust to component freezing (all variants IP $\geq$ 0.998). This complementarity is not a defect but an opportunity: a system that uses PhaseTracker for efficient standard tracking and switches to SlotAttention under detected occlusion could exceed either architecture alone.

Representation geometry. PhaseTracker embeddings ($\cos\theta, \sin\theta$, amplitude for 28 phase channels; ambient dimension 84) achieve 3.3× higher k-NN purity than SlotAttention (0.687 vs. 0.207, k=5, 3 seeds). Intrinsic dimensionality analysis reveals that PhaseTracker concentrates information on a 16-dimensional manifold (19% of ambient space), consistent with the toroidal topology $\mathbb{T}^n \subset \mathbb{R}^{2n}$ of phase representations. SlotAttention uses 47 of its 64 dimensions (73%), indicating less structured, more distributed representations. Both models show weakly negative global silhouette (−0.011 vs. −0.005), confirming that the discriminating metric is local identity coherence (k-NN purity), not global cluster separation.

Gradient flow. PhaseTracker's 2-layer architecture produces 15.4× larger gradient norms per layer than SlotAttention's 13-layer design, with zero vanishing gradients. SlotAttention exhibits partial vanishing in query projection (7.1%) and slot normalisation (11.2%) layers. The compact oscillatory architecture avoids gradient dilution, concentrating the learning signal on binding-relevant parameters.

Parameter efficiency frontier. PhaseTracker achieves IP/kParam = 0.200 vs. SlotAttention's 0.012 (16.8× more efficient). Scaling PhaseTracker to 136K parameters (PT-Large, 27× PT-Small, 1.6× SlotAttention) does **not** improve performance under stress (Supplementary Table S3), confirming that SlotAttention's advantages are architectural, not capacity-based.

Chimera as computational primitive. Chimera states (BC = 0.643, 5 sequences) in OscilloSim demonstrate that differentiable neural frameworks can reproduce complex spatiotemporal phenomena from the physics literature. Chimera strength persists across system sizes ($N \in \{128, 256, 512, 1024\}$; BC > 0.5 at all sizes). Community topologies yield more reproducible chimera dynamics than ring topology (CI width 0.047 vs. 0.151; Supplementary Table S5).

7 Discussion

A publishable and honest outcome. PhaseTracker does not beat SlotAttention on raw accuracy, but achieves practical equivalence (TOST $p = 0.004$) with 16.8× fewer parameters, 44.7× fewer FLOPs, and 143× fewer binding-specific parameters. We characterise precisely *when* and *why* each architecture fails, providing actionable guidance for system design.

Complementary, not competing. The distinct failure modes (SlotAttention: GRU-critical; PhaseTracker: occlusion-sensitive) suggest that phase binding and competitive attention are complementary mechanisms. A system that uses PhaseTracker for efficient standard tracking and switches to SlotAttention under detected occlusion could exceed either architecture alone. The 143× binding parameter gap (Proposition 2) means PhaseTracker handles the common case at negligible cost, with SlotAttention reserved for challenging frames.

Connection to the temporal correlation hypothesis. PhaseTracker provides a computational instantiation of the temporal correlation hypothesis (TCH) in a modern deep learning framework. We identify 11 parameter-level correspondences between PhaseTracker and the neuroscience literature: coupling matrices ↔ synaptic connectivity; natural frequencies ↔ intrinsic neural oscillation; PAC gating ↔ theta-gamma cross-frequency coupling; Stuart–Landau amplitude ↔ cortical gain control [4, 6, 7, 12, 15]. This mapping is interpretive, not a biological model—PhaseTracker simulates millisecond temporal dynamics on a discrete computer, not cortical circuits. However, the success of phase-based binding in achieving near-optimal tracking with extreme parameter efficiency lends computational credibility to the TCH: synchronisation-based binding *works* in a regime where attention-based binding works equally well, while using qualitatively fewer resources.

Supercritical regime and self-healing. The 15–207× supercritical operating margin (Proposition 1) explains both the architectural robustness (no single component is critical) and the fast occlusion recovery ($\tau_r < 1$ frame). From a dynamical systems perspective, training shapes the energy landscape to have a deep, wide basin around the phase-locked state. This stands in contrast to SlotAttention's GRU-based memory, which provides explicit state persistence but creates a critical single-point-of-failure dependency.

Comparison with AKOrN. AKOrN [11] adds Kuramoto dynamics to self-attention for object discovery, operating on S^{d-1} (high-dimensional phase). PhaseTracker uses scalar phases in a neuroscience-grounded multi-band hierarchy, applied to temporal tracking. The two approaches are complementary: AKOrN demonstrates that oscillatory dynamics improve *static* object decomposition; PhaseTracker demonstrates that they provide *temporal* binding at extreme parameter efficiency.

Limitations. PhaseTracker's occlusion sensitivity (IP = 0.416 at 60%), velocity degradation (IP = 0.899 at 10×), and noise fragility ($\sigma_c = 4.17$) are significant weaknesses. All experiments use

synthetic data; generalisation to natural video benchmarks (MOTChallenge, DAVIS) remains future work. The theoretical convergence bound (Proposition 1) uses the all-to-all Kuramoto threshold, which is vacuous for PhaseTracker's sparse topology; a tighter spectral bound is needed. Our RTX 4060 hardware limits the largest system sizes explored.

8 Conclusion

PRINet demonstrates that oscillatory dynamics provide a viable, parameter-efficient mechanism for temporal object binding. Trained PhaseTracker (4,991 parameters) achieves practical equivalence to SlotAttention (83,904 parameters)—TOST equivalence at $\delta = 0.5\%$ ($p = 0.004$)—through continuous phase coherence rather than learned gated recurrence, using 16.8× fewer parameters, 44.7× fewer FLOPs, and 143× fewer binding-specific parameters.

The two architectures exhibit complementary failure modes: SlotAttention excels under occlusion and adversarial perturbation while PhaseTracker offers intrinsic temporal coherence, component robustness, a compact 16-dimensional manifold representation (3.3× higher k-NN purity), and operation deep within the supercritical Kuramoto regime ($K_{\text{eff}}/K_c = 15$–$207\times$).

We confirm chimera states (BC $= 0.643 \pm 0.057$, 5 sequences) within OscilloSim v2.0, reproducing Abrams & Strogatz (2004) in a differentiable neural framework scaling to 1M+ oscillators.

The v3.0.0 release (1,640 tests, 175 public API symbols, 152 benchmark artefacts, OscilloSim scaling to 1M+ oscillators) and `reproduce.py` establish a fully reproducible foundation for oscillatory-attentive hybrid architectures.

Reproducibility Statement

All 152 benchmark JSON artefacts, trained model checkpoints, and pre-registration hashes (SHA-256) are included in supplementary materials. Running `reproduce.py` regenerates all figures and tables from stored artefacts. Source code for PRINet v3.0.0 is available at Symbo-gif/PRINet-3.0.0.

Compute budget: All experiments ran on a single NVIDIA RTX 4060 (8 GB VRAM), AMD Ryzen 8c/16t, 32 GB RAM, under 24 total GPU-hours across 4 experimental phases.

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